CHAPTER

03

Fluid Mechanics

A **fluid** is a substance that **flows**: it continually deforms under a shear (sideways) force. Liquids and gases are both fluids. **Fluid statics** studies fluids at rest (pressure, Pascal, Archimedes); **fluid dynamics** studies fluids in motion (continuity, Bernoulli). We begin with how solids respond to forces (**elasticity**), because the same ideas of stress, strain and modulus lead to the bulk modulus of fluids. Applications are everywhere: blood flow, breathing, hydraulic brakes, ships, aeroplanes, dams and water pipes.

LESSONS IN THIS CHAPTER

3.1	Properties of Bulk Matter: Stress, Strain and Elasticity
3.2	Density and Pressure in Static Fluids
3.3	Pascal's Principle, Buoyancy and Archimedes' Principle
3.4	Fluid Dynamics: Continuity and Bernoulli's Equation

Freshman A+ · General Physics (Phys 1011)

CHAPTER 3 · LESSON 3.1

Properties of Bulk Matter: Stress, Strain and Elasticity

AFTER THIS LESSON YOU CAN

- Distinguish elastic and plastic behaviour
- Define tensile, shear and volume stress and strain
- Use Young's, shear and bulk moduli to solve problems
- Calculate strain energy

3.1.1 Elastic and plastic behaviour

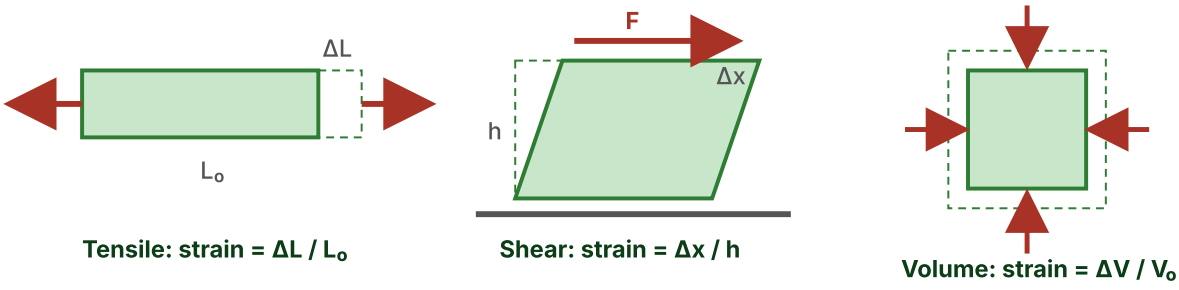
KEY DEFINITIONS

- **Elastic material:** regains its original shape and size when the deforming force is removed (a rubber band, a steel spring within its limit).
- **Elastic deformation:** a **reversible** deformation produced by a force within the **elastic limit**.
- **Plastic deformation:** a **permanent**, irreversible deformation produced when a force exceeds the elastic limit.
- **Plastic material:** does **not** regain its original shape and size (clay, putty, chewing gum).

3.1.2 Stress and strain

$$\text{Stress} = \frac{F}{A} \ \ (\text{unit: N/m}^2 = \text{Pa}) \qquad \text{Strain} = \frac{\text{change in size}}{\text{original size}} \ \ (\text{no unit})$$

Stress is the force per unit area that causes a deformation; **strain** measures how much deformation results. Strain is a **ratio**, so it is **dimensionless**.



Tensile: strain = ΔL / L_o **Shear: strain = Δx / h** **Volume: strain = ΔV / V_o**

The three kinds of deformation

3.1.3 The three elastic moduli

For small deformations, **stress is proportional to strain** (Hooke's law for materials):

$$\text{Elastic modulus} = \frac{\text{stress}}{\text{strain}} \qquad (\text{unit: Pa})$$

THE THREE MODULI

Modulus	Deformation	Formula	Measures resistance to
Young's modulus Y	stretching / compressing a rod or wire	$Y = \dfrac{F/A}{\Delta L/L_0}$	change of **length**
Shear modulus S	sideways force on one face	$S = \dfrac{F/A}{\Delta x/h}$	sliding of layers (change of **shape**)
Bulk modulus B	uniform pressure from all sides	$B = -\dfrac{\Delta P}{\Delta V/V_0}$	change of **volume**

- **Compressibility** $= 1/B$: a large bulk modulus means the material is hard to compress.
- The **minus sign** in B makes it positive: an increase in pressure ($\Delta P > 0$) decreases the volume ($\Delta V < 0$).
- **Liquids have a bulk modulus, but no Young's or shear modulus**: a liquid cannot sustain a tensile or shear stress, it simply flows.
- **Strain energy** stored in a stretched wire: $U = \frac{1}{2} F\,\Delta L = \frac{1}{2} \cdot \dfrac{YA}{L_0}(\Delta L)^2$.

Typical values: steel $Y \approx 20 \times 10^{10}$ Pa, aluminium 7.0×10^{10} Pa, brass 9.1×10^{10} Pa (bulk 6.1×10^{10} Pa), water $B \approx 0.21 \times 10^{10}$ Pa.

WORKED EXAMPLE 1: STEEL WIRE FOR A STAGE ACTOR (MODULE)

A 10 m steel wire ($Y = 20 \times 10^{10}$ Pa) carries a tension of 940 N. What diameter keeps its stretch below 0.5 cm?

$A = \dfrac{FL_0}{Y\,\Delta L} = \dfrac{(940)(10)}{(20 \times 10^{10})(0.005)} = 9.4 \times 10^{-6}$ m².

$r = \sqrt{A/\pi} = 1.73 \times 10^{-3}$ m → $d = 2r \approx 3.5$ mm.

WORKED EXAMPLE 2: BRASS SPHERE IN THE OCEAN (MODULE ACTIVITY)

A 0.50 m³ brass sphere ($B = 6.1 \times 10^{10}$ Pa) is lowered from air (1.0×10^5 Pa) to a depth where the pressure is 2.0×10^7 Pa. Find the change in volume.

$\Delta V = -\dfrac{V_0\,\Delta P}{B} = -\dfrac{(0.50)(2.0 \times 10^7 - 1.0 \times 10^5)}{6.1 \times 10^{10}} = -1.6 \times 10^{-4}$ m³ (it shrinks by 160 cm³).

WORKED EXAMPLE 3: A LOAD ON A WIRE (MODULE PROBLEM 1)

A 200 kg load hangs on a 4.00 m wire of cross-section 0.200×10^{-4} m² and $Y = 8.00 \times 10^{10}$ Pa ($g = 9.8$). Find the stretch.

$\Delta L = \dfrac{FL_0}{AY} = \dfrac{(1960)(4.00)}{(0.200 \times 10^{-4})(8.00 \times 10^{10})} = 4.9 \times 10^{-3}$ m = 4.9 mm.

! ELASTICITY TRAPS

- Stress has the **same unit as pressure** (Pa); strain has **no unit**.
- For a given force, the stretch is **proportional to the length** and **inversely proportional to the area**: a wire twice as long stretches twice as much; a wire with twice the diameter (4 × the area) stretches ¼ as much.
- The modulus depends on the **material**, not on the size of the sample.

Exam Practice

6 QUESTIONS · LESSON 3.1

1 Arba Minch University · Mid Exam

Define: (a) elastic material, (b) plastic material, (c) elastic modulus.

2 Model question

Which of the following has no unit?

A) Strain
B) Stress
C) Young's modulus
D) Bulk modulus

3 Serway / OpenStax-style problem

Why is no Young's modulus or shear modulus given for water?

A) Water is incompressible
B) A liquid cannot sustain a tensile or shear stress; it flows
C) Water has no density
D) The values are too small to measure

4 Serway / OpenStax-style problem

Two wires of the same material and length carry the same load. Wire B has twice the diameter of wire A. The stretch of B compared with A is:

A) twice as large
B) the same
C) half as large
D) one quarter as large

5 Model question (module problem)

When water freezes it expands by about 9.0 %. What pressure increase would occur inside an engine block if the water in it froze? (Bulk modulus of ice $= 2.0 \times 10^9$ Pa.)

6 Serway / OpenStax-style problem

Steel ruptures if the shear stress exceeds 4.0×10^8 Pa. Find the force needed to (a) shear a steel bolt of diameter 1.00 cm, and (b) punch a 1.00 cm diameter hole in a steel plate 0.500 cm thick.

Answers & Explanations

1. **See explanation**

(a) A material that regains its original shape and size when the deforming force is removed.
(b) A material that does not regain its original shape and size when the force is removed.
(c) The ratio of stress to strain, $\text{modulus} = \text{stress}/\text{strain}$ (unit Pa); it measures a material's resistance to deformation.

2. **A**

Strain

Strain is a ratio of two lengths (or two volumes), so it is dimensionless. Stress and all the moduli are measured in pascals.

3. **B**

A liquid cannot sustain a tensile or shear stress; it flows

A fluid responds to a shear or tensile force by flowing, so these moduli are not defined. Liquids do have a bulk modulus.

4. **D**

one quarter as large

$\Delta L = FL_0/(AY)$ and $A \propto d^2$. Doubling d makes A four times larger, so ΔL is ¼.

5. **1.8×10^8 Pa**

To squeeze the ice back into the original volume, $|\Delta V/V| = 0.090$: $\Delta P = B\,|\Delta V/V| = (2.0 \times 10^9)(0.090) = 1.8 \times 10^8$ Pa (about 1800 atm, which is why engine blocks crack).

6. **(a) 3.1×10^4 N; (b) 6.3×10^4 N**

(a) Sheared area = cross-section $= \pi(0.0050)^2 = 7.85 \times 10^{-5}$ m² → $F = 4.0 \times 10^8 \times 7.85 \times 10^{-5} = 3.1 \times 10^4$ N.
(b) Sheared area = the curved side of the hole $= \pi d\,t = \pi(0.0100)(0.00500) = 1.57 \times 10^{-4}$ m² → $F = 6.3 \times 10^4$ N.

CHAPTER 3 · LESSON 3.2

Density and Pressure in Static Fluids

AFTER THIS LESSON YOU CAN

- Calculate density, weight density and specific gravity
- Compute pressure at a depth
- Distinguish absolute and gauge pressure
- Understand atmospheric pressure and the barometer

3.2.1 Density and specific gravity

D DEFINITIONS

- **Density:** $\rho = \dfrac{m}{V}$ (SI unit kg/m³). Water: $\rho_w = 1000$ kg/m³ = 1 g/cm³ = 1 kg/L.
- **Weight density:** $\rho g = \dfrac{W}{V}$ (N/m³).
- **Specific gravity (relative density):** $SG = \dfrac{\rho}{\rho_{\text{water}}}$, the ratio to the density of water at 4 °C. It is **dimensionless**. Aluminium: $\rho = 2700$ kg/m³, so $SG = 2.7$.

✎ WORKED EXAMPLE 1: A WOODEN SPHERE (MODULE)

A wooden sphere of radius 0.10 m has a mass of 1.0 kg. Find its density and specific gravity.
$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(0.10)^3 = 4.19 \times 10^{-3}$ m³ → $\rho = 1.0/4.19 \times 10^{-3} = 239$ kg/m³; $SG = 0.239$ (it floats on water).

3.2.2 Pressure

D PRESSURE

The **perpendicular force per unit area**:

$$P = \frac{F_\perp}{A} \qquad \text{unit: pascal, } 1\ \text{Pa} = 1\ \text{N/m}^2$$

Pressure is a **scalar**: at a point in a fluid it acts **equally in all directions**. 1 atm = 1.013×10^5 Pa = 101.3 kPa = 760 mmHg.

3.2.3 Pressure at a depth

The weight of the fluid above presses down, so pressure increases with depth:

$$P = P_0 + \rho g h$$

where P_0 is the pressure at the surface (usually atmospheric) and h is the depth.

★ THREE FACTS ABOUT PRESSURE IN A FLUID AT REST

1. **All points at the same level** in the same fluid have the **same pressure**.
2. Pressure **increases with depth** (linearly, for a liquid of constant density).
3. Pressure **does not depend on the shape of the container** (the "hydrostatic paradox"): a narrow tube and a wide tank filled to the same height have the same pressure at the bottom.

D GAUGE AND ABSOLUTE PRESSURE

- **Gauge pressure:** pressure **above** atmospheric, $P_g = P - P_{\text{atm}} = \rho g h$. A tyre gauge reads this.
- **Absolute pressure:** the total, $P = P_{\text{atm}} + P_g$.
- Absolute pressure cannot be negative (fluids push, they do not pull), so the smallest possible gauge pressure is $-P_{\text{atm}}$ (a perfect vacuum).

E ATMOSPHERIC PRESSURE AND THE BAROMETER

Atmospheric pressure is caused by the weight of the air above us. It **decreases with altitude** because the air becomes thinner (less dense). A mercury barometer balances the atmosphere with a column of mercury: $P_{\text{atm}} = \rho_{Hg} g h$, giving $h = 760$ mm at sea level. A water barometer would need a column of about 10.3 m.

✎ WORKED EXAMPLE 2: SCUBA DIVER (MODULE)

A wreck lies 18.3 m below the sea surface ($\rho = 1025$ kg/m³, $g = 9.8$). Find the gauge and absolute pressure on a diver there.

$P_g = \rho g h = (1025)(9.8)(18.3) = 1.84 \times 10^5$ Pa.

$P = 1.013 \times 10^5 + 1.84 \times 10^5 = 2.85 \times 10^5$ Pa (about 2.8 atm).

! PRESSURE TRAPS

- In $P = \rho g h$, h is the **vertical depth** below the surface, not the distance along a slanted tube.
- Pressure at the bottom of a container depends on **depth and density only**, not on the total amount of liquid or the area.
- Watch the units: convert cm to m and g/cm³ to kg/m³ (× 1000).
- "Newton per cubic kilometre" is **not** a unit of density; it is a weight density (force per volume). But Pa/(m²/s²) **is** a density unit: $\dfrac{\text{kg m}^{-1}\text{s}^{-2}}{\text{m}^2\text{s}^{-2}} = \text{kg/m}^3$.

Exam Practice

6 QUESTIONS · LESSON 3.2

1 Arba Minch University · Mid Exam

Which of the following is NOT a unit of density?

A) kg/litre
B) gram/(mm)³
C) newton/(km)³
D) pascal/(m²/s²)

2 Arba Minch University · Mid Exam

Which of the following is NOT correct about fluid pressure?

A) All points at the same level in a fluid have the same pressure
B) Fluid pressure increases with increasing depth
C) Fluid pressure does not depend on the shape of the container
D) All points at the same level in a fluid have different pressures

3 Arsi University · Final Exam 2014

Which of the following statements about fluids at rest are true? (i) All points at the same level have the same pressure. (ii) Pressure increases with depth. (iii) Pressure does not depend on the shape of the container.

A) (i) only
B) (i) and (ii) only
C) (ii) and (iii) only
D) All

4 Model question

The gauge pressure at the bottom of a glass filled with water to a depth of 30 cm is ($g = 9.8$ m/s²):

A) 294 Pa
B) 3.0 × 10^4 Pa
C) 1.04 × 10^5 Pa
D) 2940 Pa

5 Model question (module problem)

Calculate the absolute pressure at an ocean depth of 1.0×10^3 m ($\rho = 1.025 \times 10^3$ kg/m³, $g = 9.8$ m/s²).

6 Serway / OpenStax-style problem

Why does atmospheric pressure decrease as you climb a mountain?

A) Gravity is much weaker at the top
B) There is less air above you and the air is less dense
C) The temperature rises with altitude
D) Air pressure acts only downward

Answers & Explanations

1. C

newton/(km)³

Newton per volume is a **weight density** (force/volume). Pa/(m²/s²) reduces to kg/m³, so D **is** a density unit.

2. D

All points at the same level in a fluid have different pressures

A, B and C are the three basic facts of hydrostatics; D contradicts A.

3. D

All

All three are properties of a static fluid: $P = P_0 + \rho g h$ depends only on depth.

4. D

2940 Pa

$P_g = \rho g h = 1000 \times 9.8 \times 0.30 = 2940$ Pa. (The absolute pressure is $1.013 \times 10^5 + 2940 = 1.04 \times 10^5$ Pa.)

5. **1.02 × 10^7 Pa (about 100 atm)**

$P = P_{\text{atm}} + \rho g h = 1.013 \times 10^5 + (1025)(9.8)(1000) = 1.013 \times 10^5 + 1.005 \times 10^7 = 1.02 \times 10^7$ Pa.

6. B

There is less air above you and the air is less dense

Atmospheric pressure is the weight of the air column above; higher up there is less air, and it is thinner.

CHAPTER 3 · LESSON 3.3

Pascal's Principle, Buoyancy and Archimedes' Principle

AFTER THIS LESSON YOU CAN

- State Pascal's principle and analyse hydraulic systems
- State Archimedes' principle and explain the origin of the buoyant force
- Predict whether an object floats or sinks
- Find density from weighing in air and in a liquid

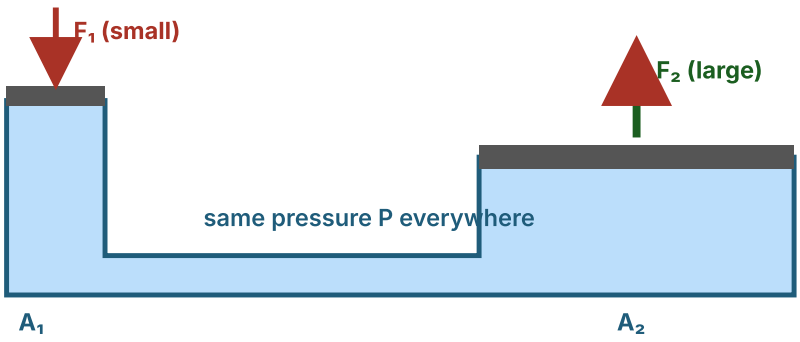
3.3.1 Pascal's principle

D PASCAL'S PRINCIPLE

Pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the walls of its container.

The hydraulic lift (press): a small force F_1 on a small piston A_1 creates a pressure F_1/A_1, which reaches the large piston A_2 unchanged:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \Longrightarrow \quad F_2 = F_1 \frac{A_2}{A_1}$$



A hydraulic lift multiplies force by A_2/A_1

★ FORCE IS MULTIPLIED, ENERGY IS NOT

The output force is larger by the factor A_2/A_1, but the large piston moves a **smaller distance**: $A_1 d_1 = A_2 d_2$ (the same volume of fluid moves). So the **work in = work out**: $F_1 d_1 = F_2 d_2$. A hydraulic machine is a force multiplier, not an energy source. Applications: car lifts, hydraulic **brakes**, jacks, excavators, barber chairs.

✎ WORKED EXAMPLE 1: HYDRAULIC LIFT (MODULE AND ARBA MINCH EXAM)

The small piston has area 0.2 m² and the large piston 0.9 m². A car weighing 1.2×10^4 N sits on the large piston. What force on the small piston supports it?

$F_1 = F_2 \dfrac{A_1}{A_2} = (1.2 \times 10^4)\dfrac{0.2}{0.9} = 2.7 \times 10^3$ N.

3.3.2 Buoyancy and Archimedes' principle

D ARCHIMEDES' PRINCIPLE

Any body completely or partially submerged in a fluid is buoyed up by a force equal to the weight of the fluid displaced by the body.

$$F_B = W_{\text{displaced fluid}} = \rho_{\text{fluid}}\, V_{\text{submerged}}\, g$$

Why it happens: pressure increases with depth, so the fluid pushes **up on the bottom** of the object harder than it pushes **down on the top**. The difference is the upward buoyant force. The sideways pushes cancel, so the net horizontal force on a submerged object is zero.

★ THE BUOYANT FORCE DEPENDS ONLY ON THE DISPLACED VOLUME AND THE FLUID

F_B depends on the **density of the fluid** and the **submerged volume**, **not** on the object's own density, mass or depth (once fully submerged). A lead block and an iron block of the **same volume**, both fully submerged, feel the **same buoyant force**.

Floating and sinking

Condition	Result
$\rho_{\text{object}} > \rho_{\text{fluid}}$	**sinks** ($W > F_B$ even when fully submerged)
$\rho_{\text{object}} = \rho_{\text{fluid}}$	stays suspended at any depth (fish, submarines adjust this)
$\rho_{\text{object}} < \rho_{\text{fluid}}$	**floats**, partly submerged, with $F_B = W$

F FLOATING OBJECT

$$\frac{V_{\text{submerged}}}{V_{\text{total}}} = \frac{\rho_{\text{object}}}{\rho_{\text{fluid}}}$$

Ice (917 kg/m³) in sea water (1025 kg/m³): $917/1025 = 0.89$, so about **89 % of an iceberg is under water**. A steel ship floats because its hollow hull displaces a volume of water whose weight equals the ship's weight.

F WEIGHING IN AIR AND IN A LIQUID

- **Apparent weight** $= W - F_B$, so the buoyant force is the **loss of weight**: $F_B = W_{\text{air}} - W_{\text{liquid}}$.
- Density of a sinking object: $\dfrac{\rho_{\text{object}}}{\rho_{\text{liquid}}} = \dfrac{W_{\text{air}}}{W_{\text{air}} - W_{\text{liquid}}}$

✎ WORKED EXAMPLE 2: UNKNOWN MATERIAL (MODULE)

A sample weighs 300 N in air and 200 N in alcohol ($\rho = 790$ kg/m³). Find its density.

$F_B = 300 - 200 = 100$ N → $\rho = \frac{300}{100} \times 790 = 2.37 \times 10^3$ kg/m³.

WORKED EXAMPLE 3: IS THE BRACELET PURE SILVER? (AASTU EXAM)

A bracelet has a mass of 10 g in air and an apparent mass of 9 g in water. Silver has density 10.5 g/cm³. Is it pure silver?

Lost mass = mass of displaced water = 1 g → $V = 1$ cm³ (water is 1 g/cm³).

$\rho = 10\ \text{g}/1\ \text{cm}^3 = 10$ g/cm³ < 10.5 g/cm³ → **not pure silver** (it is mixed with a lighter metal).

BUOYANCY TRAPS

- F_B uses the density of the **fluid**, not of the object.
- For a **floating** object, F_B = its **full weight**; for a sunken one, F_B < weight.
- A boat floating in fresh water rises slightly when it sails into sea water (denser water: less volume needs to be displaced), but the buoyant force is the **same** (it equals the boat's weight in both).
- The buoyant force on a fully submerged object does not increase with depth (liquids are nearly incompressible).

Exam Practice

9 QUESTIONS · LESSON 3.3

1 Arba Minch University · Mid Exam

The upward force acting on an object submerged in a fluid is called:

A) Drag **B)** Weight
C) Pressure **D)** Buoyant force

2 Haramaya University · Mid Exam

Two objects of different density but the same volume, both denser than water, can experience equal buoyant forces when immersed in water.

True / False

3 Arba Minch University · Mid Exam

The small piston of a hydraulic lift has an area of 0.2 m² and the large piston 0.9 m². A car weighing 1.2×10^4 N sits on the large piston. The force needed on the small piston is about:

A) 5.4×10^4 N **B)** 2.7×10^3 N
C) 1.2×10^4 N **D)** 6.0×10^2 N

4 Hawassa University · Mid Exam

In a hydraulic lift, the small piston has an area of 2 cm² and the large piston 20 cm². What force on the small piston lifts a 400 N load?

5 Bahir Dar University · Mid Exam

State Archimedes' principle.

6 AASTU · Final Exam

A bracelet has a mass of 10 g in air and an apparent mass of 9 g when fully submerged in water. Is it pure silver ($\rho_{Ag} = 10.5$ g/cm³)?

7 Model question (module)

Lead is denser than iron. A lead block and an iron block of the same volume are fully submerged in water. The buoyant force on the lead block is:

A) greater **B)** smaller
C) zero **D)** the same

8 Serway / OpenStax-style problem

A wooden block floats in water with 60 % of its volume submerged. Its density is:

A) 600 kg/m³ **B)** 400 kg/m³
C) 1000 kg/m³ **D)** 1667 kg/m³

9 Model question (module)

A sample weighs 300 N in air and 200 N when submerged in an alcohol of density 790 kg/m³. Find (a) the buoyant force, (b) the volume of the sample, and (c) its density ($g = 9.8$).

Answers & Explanations

1. **D**

Buoyant force

By Archimedes' principle, the fluid exerts an upward buoyant force equal to the weight of the fluid displaced.

2. **True**

Both sink and are fully submerged, so they displace the same volume of water: $F_B = \rho_w V g$ is the same. The object's own density does not enter.

3. **B**

2.7×10^3 **N**

$F_1 = F_2 A_1 / A_2 = 1.2 \times 10^4 \times 0.2/0.9 = 2.7 \times 10^3$ N.

4. **40 N**

$$F_1 = F_2 \frac{A_1}{A_2} = 400 \times \frac{2}{20} = 40 \text{ N}.$$

5. **See explanation**

Any body completely or partially submerged in a fluid is buoyed up by a force equal to the weight of the fluid it displaces: $F_B = \rho_{\text{fluid}} V_{\text{sub}} g$.

6. **No: its density is 10 g/cm³**

Displaced water = 1 g → $V = 1$ cm³. Density $= 10/1 = 10$ g/cm³, less than 10.5 g/cm³.

7. **D**

the same

$F_B = \rho_w V g$ depends only on the displaced volume, which is the same.

8. **A**

600 kg/m³

$V_{\text{sub}}/V = \rho_{\text{obj}}/\rho_w \Rightarrow \rho_{\text{obj}} = 0.60 \times 1000 = 600$ kg/m³.

9. **(a) 100 N; (b) 0.0129 m³; (c) 2.37 × 10³ kg/m³**

(a) $F_B = 300 - 200 = 100$ N.
(b) $V = F_B/(\rho g) = 100/(790 \times 9.8) = 1.29 \times 10^{-2}$ m³.
(c) $\rho = m/V = (300/9.8)/0.0129 = 2.37 \times 10^3$ kg/m³.

CHAPTER 3 · LESSON 3.4

Fluid Dynamics: Continuity and Bernoulli's Equation

AFTER THIS LESSON YOU CAN

- Describe ideal fluid flow and distinguish laminar and turbulent flow
- Apply the equation of continuity
- Apply Bernoulli's equation and explain its everyday consequences
- Use Torricelli's theorem

3.4.1 Ideal fluid flow

D LAMINAR (STREAMLINE) FLOW

Smooth flow in regular layers. Every particle passing a point follows the same path (a **streamline**) as the ones before it. The velocity at each point stays constant in time.

! TURBULENT FLOW

Irregular, chaotic flow with eddies, above a certain speed. Energy is lost to internal friction between layers, called **viscosity**.

★ THE FOUR ASSUMPTIONS OF AN IDEAL FLUID

1. **Non-viscous:** no internal friction between layers.
2. **Steady:** the velocity at each point does not change with time.
3. **Incompressible:** the density is constant.
4. **Irrotational:** the fluid has no angular momentum about any point (no swirls).

Bernoulli's equation is valid only for flows that satisfy these conditions, so it is **valid for irrotational flow**, not for viscous, compressible or unsteady flow.

3.4.2 The equation of continuity

Conservation of mass: in a pipe with no leaks, the volume entering one end per second equals the volume leaving the other end:

$$A_1 v_1 = A_2 v_2 = Q \ (\text{constant})$$

$Q = Av$ is the **volume flow rate** (m³/s). For circular pipes, $v_2 = v_1 \left(\frac{d_1}{d_2}\right)^2$.

E EVERYDAY CONTINUITY

Put your thumb over the end of a garden hose: the area falls, so the water comes out **faster**. Rivers flow faster where they are narrow. Blood flows slowly in the capillaries because their **total** cross-sectional area is huge.

3.4.3 Bernoulli's equation

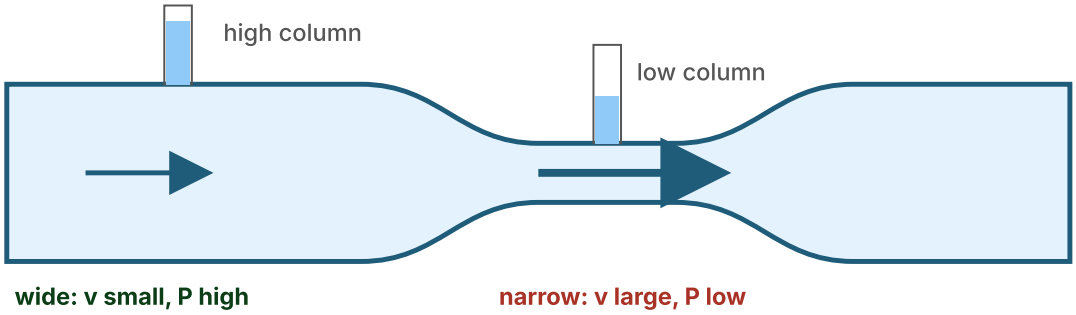
Conservation of energy for a flowing ideal fluid, per unit volume, along a streamline:

$$P_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2$$

Term	Meaning
P	pressure (work done by the surrounding fluid)
$\tfrac{1}{2}\rho v^2$	kinetic energy per unit volume
$\rho g y$	gravitational potential energy per unit volume

D **BERNOULLI'S PRINCIPLE**

For a horizontal flow ($y_1 = y_2$): **where the fluid moves faster, its pressure is lower.**



Venturi tube: continuity makes the fluid speed up in the throat, and Bernoulli says its pressure drops there

E **BERNOULLI IN DAILY LIFE**

- **Aeroplane wing:** air flows faster over the curved top surface, so the pressure on top is lower than underneath → **lift**.
- A strong wind passing over a roof lowers the pressure above it, so roofs can be **lifted off** in storms.
- A **curve ball** (spinning ball), a **sailboat** sailing into the wind, smoke rising up a **chimney**, an atomiser or perfume spray, a carburettor, and two papers moving **together** when you blow between them.

F **TORRICELLI'S THEOREM**

Water flows out of a small hole a depth h below the surface of an open tank with speed

$$v = \sqrt{2gh}$$

the same as an object falling freely from height h (Bernoulli with $P_1 = P_2 = P_{\text{atm}}$ and $v_1 \approx 0$).

✎ **WORKED EXAMPLE 1: HOT-WATER HEATING SYSTEM (MODULE AND GONDAR EXAM)**

Water is pumped at 0.50 m/s through a 4.0 cm diameter pipe in the basement at 3.03×10^5 Pa. Find the speed and pressure in a 2.6 cm diameter pipe on the second floor, 5.0 m higher.

Continuity: $v_2 = v_1\left(\frac{d_1}{d_2}\right)^2 = 0.50\left(\frac{4.0}{2.6}\right)^2 = 1.18$ m/s.
Bernoulli: $P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2) - \rho g(y_2 - y_1)$
$= 3.03 \times 10^5 + \frac{1}{2}(1000)(0.25 - 1.40) - (1000)(9.8)(5.0)$
$= 3.03 \times 10^5 - 575 - 4.9 \times 10^4 = 2.53 \times 10^5$ Pa.

WORKED EXAMPLE 2: GARDEN HOSE (AASTU EXAM)

A hose of radius 1.0 cm fills a 15 L bucket in 1.0 min. (a) Find the water speed. (b) If you cover half the opening with your thumb, what is the new speed?
$Q = 15 \times 10^{-3}/60 = 2.5 \times 10^{-4}$ m³/s; $A = \pi(0.010)^2 = 3.14 \times 10^{-4}$ m².
(a) $v_1 = Q/A = 0.80$ m/s. (b) Half the area → $v_2 = 2v_1 = 1.6$ m/s.

FLUID-DYNAMICS TRAPS

- Continuity: **smaller area → larger speed**. Halving the **diameter** makes the speed **4 times** larger.
- Bernoulli: **faster → lower pressure**, the opposite of most students' intuition.
- Don't forget the height term $\rho g y$ when the pipe goes up or down.
- Use **absolute** or **gauge** pressure consistently on both sides.

Exam Practice

8 QUESTIONS · LESSON 3.4

1 Arba Minch University · Mid Exam

Bernoulli's equation in fluid dynamics is valid for:

A) Compressible flows
B) Unsteady flows
C) Irrotational flows
D) Viscous flows

2 Arba Minch University · Mid Exam

State the equation of continuity for a fluid.

3 Model question

Water flows through a horizontal pipe whose diameter halves at a constriction. The speed in the narrow section is:

A) four times as large
B) twice as large
C) the same
D) half as large

4 Model question

In the narrow section of the pipe in the previous question, the pressure is:

A) higher than in the wide section
B) zero
C) the same
D) lower than in the wide section

5 Serway / OpenStax-style problem

The lift on an aeroplane wing is best explained by:

A) Bernoulli's principle
B) Archimedes' principle
C) Pascal's principle
D) Hooke's law

6 University of Gondar · Final Exam

Water is pumped at 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.03×10^5 Pa. Find the velocity and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above.

7 AASTU · Final Exam

Water from a hose of radius 1.0 cm fills a 15 L container in 1.0 min. (a) What is the speed of the water leaving the hose? (b) If half the opening is blocked, what is the new speed?

8 Serway / OpenStax-style problem

A large open water tank has a small hole 5.0 m below the water surface ($g = 9.8$). Find the speed of the water leaving the hole, and the flow rate if the hole's area is 2.0 cm².

Answers & Explanations

1. C

Irrotational flows

Bernoulli's equation assumes an ideal fluid: incompressible, steady, non-viscous and irrotational.

2. $A_1v_1 = A_2v_2$

For an incompressible fluid in a pipe with no leaks, the volume flow rate $Q = Av$ is the same at every cross-section (conservation of mass).

3. A

four times as large

$v_2 = v_1(d_1/d_2)^2 = v_1 \times 2^2 = 4v_1.$

4. D

lower than in the wide section

Bernoulli: for a horizontal pipe, faster flow means lower pressure.

5. A

Bernoulli's principle

The air moves faster over the top of the wing, so the pressure there is lower than below; the pressure difference produces an upward force.

6. **1.18 m/s; 2.53 × 10⁵ Pa**

$v_2 = 0.50(4.0/2.6)^2 = 1.18$ m/s.

$P_2 = 3.03\times10^5 + 500(0.25-1.40) - 1000(9.8)(5.0) = 2.53\times10^5$ Pa.

7. **(a) 0.80 m/s; (b) 1.6 m/s**

(a) $v = Q/A = (2.5\times10^{-4})/(3.14\times10^{-4}) = 0.80$ m/s.

(b) The flow rate is unchanged and the area halves, so v doubles: 1.6 m/s.

8. **9.9 m/s; 2.0 × 10⁻³ m³/s (2.0 L/s)**

Torricelli: $v = \sqrt{2gh} = \sqrt{2(9.8)(5.0)} = 9.9$ m/s. $Q = Av = (2.0\times10^{-4})(9.9) = 1.98\times10^{-3}$ m³/s.

CHAPTER 3 · QUICK REVISION

Fluid Mechanics — at a glance

CHAPTER 3 FORMULA SHEET

Topic	Formulas
Stress, strain	stress $= F/A$ (Pa); strain $= \Delta L/L_0$, $\Delta x/h$, $\Delta V/V_0$ (no unit)
Moduli	$Y = \dfrac{F/A}{\Delta L/L_0}$, $S = \dfrac{F/A}{\Delta x/h}$, $B = -\dfrac{\Delta P}{\Delta V/V_0}$
Density	$\rho = m/V$; $SG = \rho/\rho_w$; $\rho_w = 1000$ kg/m³
Pressure	$P = F/A$; $P = P_0 + \rho g h$; $P_{\text{abs}} = P_{\text{atm}} + P_g$; 1 atm $= 1.013 \times 10^5$ Pa
Pascal	$F_1/A_1 = F_2/A_2$; $A_1 d_1 = A_2 d_2$
Archimedes	$F_B = \rho_f V_{\text{sub}} g$; floating: $V_{\text{sub}}/V = \rho_{\text{obj}}/\rho_f$
Continuity	$A_1 v_1 = A_2 v_2 = Q$
Bernoulli	$P + \frac{1}{2}\rho v^2 + \rho g y =$ constant; Torricelli $v = \sqrt{2gh}$

THE FOUR PRINCIPLES IN ONE LINE EACH

Pascal: pressure spreads equally (hydraulics). **A**rchimedes: up-push = weight of fluid displaced. **C**ontinuity: narrow → fast. **B**ernoulli: fast → low pressure. ("**PACB**: Push, Archimedes, Continuity, Bernoulli")